

Q1 #1)

$$m(a + bx) = a + b \times m(x)$$

$$m(x) = \frac{1}{N} \sum_{i=1}^N x_i$$

$$m(a + bx) = \frac{1}{N} \sum_{i=1}^N a + bx$$

$$\frac{1}{N} \left(a \sum_{i=1}^N 1 + b \sum_{i=1}^N x_i \right)$$

$$\frac{1}{N} \left(aN + b \sum_{i=1}^N x_i \right)$$

$$a + \frac{b}{N} \sum_{i=1}^N x_i = a + b \frac{1}{N} \sum_{i=1}^N x_i$$

$$= a + b m(x)$$

therefore

$$m(a + bx) = a + b \times m(x)$$

Q 1 #2)

$$\text{Cov}(X, a+bY) = b \times \text{Cov}(X, Y)$$

$$\text{Cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(y_i - m(Y))$$

$$\text{Cov}(X, a+bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) (a + by_i - m(a+bY))$$

$$m(a+bY) = a + b \times m(Y)$$

$$\text{Cov}(X, \underline{a+bY}) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) (\underline{a+bY_i} - \underline{a} - b m(Y))$$

$$by_i + b m(Y)$$

$$b(y_i + m(Y))$$

$$\text{Cov}(X, a+bY) = b \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) (y_i - m(Y))$$

$$= b \text{Cov}(X, Y)$$

hwo

$$\text{Cov}(X, a+bY) = b \text{Cov}(X, Y)$$

Q1 #3)

Show $\text{Cov}(a+bx, a+bx) = b^2 \text{Cov}(x, x)$

$$\text{Cov}(x, y) = \frac{1}{N} \sum_{i=1}^N (x_i - m(x))(y_i - m(y))$$

$$\text{Cov}(a+bx, a+bx) = \frac{1}{N} \sum_{i=1}^N (a+bx_i - m(a+bx))(a+bx_i - m(a+bx))$$

$$m(a+bx) = a + bm(x)$$

$$\text{Cov}(a+bx, a+bx) = \frac{1}{N} \sum_{i=1}^N (\cancel{a} + bx_i - \cancel{a} - bm(x))(\cancel{a} + bx_i - \cancel{a} - bm(x))$$

$$= \frac{1}{N} \sum_{i=1}^N (bx_i - bm(x))(bx_i - bm(x))$$

$$b(x_i - m(x))b(x_i - m(x))$$

$$b^2 (x_i - m(x))(x_i - m(x))$$

$$= b^2 \frac{1}{N} \sum_{i=1}^N (x_i - m(x))(x_i - m(x))$$

$$\underbrace{\hspace{10em}}_{\text{Cov}(x, x)}$$

$$\hookrightarrow \frac{1}{N} \sum_{i=1}^N (x_i - m(x))^2$$

$$= S^2$$

SD

$$\text{Cov}(x, x) = S^2$$

$$= b^2 \text{Cov}(x, x)$$

$$\text{thus } \text{Cov}(a+bx, a+bx) = b^2 \text{Cov}(x, x)$$

Q1 #c)

is $m(g(x)) = g(m(x))$ always true

$$m(x) = \frac{1}{N} \sum_{i=1}^N x_i \quad g(x) = x^2$$

$$m(g(x)) = \frac{1}{N} \sum_{i=1}^N x_i^2 \quad g(m(x)) = \left(\frac{1}{N} \sum_{i=1}^N x_i \right)^2$$

$$\frac{x_1^2 + x_2^2 + x_3^2}{N}$$

$$\left(\frac{x_1 + x_2 + x_3 + \dots}{N} \right)^2$$

$$x = (1, 2, 3) \quad \frac{1^2 + 2^2 + 3^2}{3} = \frac{14}{3} \quad \left(\frac{1+2+3}{3} \right)^2 = \left(\frac{6}{3} \right)^2 = 2^2 = 4$$

$$\frac{14}{3} \neq 4$$

$$\text{so } m(g(x)) \neq g(m(x))$$

$$\text{no } m(g(x)) = g(m(x))$$

is not always true